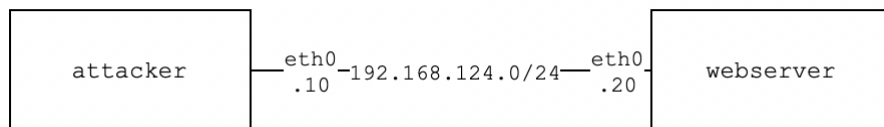


Network Security

Assignment: Transport Layer

SYN Flood [20 points]

In the topology in the image below, the webserver is vulnerable to the SYN-flooding attack.



Exploit this vulnerability to make the web server running on port 80 unavailable. Before starting with the assignment, unzip the provided file and navigate to the unzipped directory. Generate the environment using the following Docker commands.

```
docker compose build
docker compose up -d
```

To connect to the attacker container, execute the command below.

```
docker exec -ti assignment_4_syn_flood_attacker bash
```

There's a challenge: the webserver has a firewall that allows only 1 SYN packet per second from the same source. You can check whether the webserver is accessible using the following command:

```
docker exec assignment_4_syn_flood_client curl http://192.168.124.20/
```

After finishing the assignment, you can destroy the environment using the command below. Make sure your script is inside the mounted `/solution` folder, so it is saved when you destroy the environment.

```
docker compose down
```

Goal

Produce a Python3 script that accepts two arguments: the destination address and the destination port. The script should be invoked as follows:

```
python3 syn_flood.py 192.168.124.20 80
```

The script should continuously run until it is terminated. It doesn't have to generate any output. The assignment is successful if the client can no longer connect to the web server.

Submission Instructions

Your script should run on the *attacker* container. You may not use external libraries or tools different from those installed on the provided *attacker* container.